14 (a) Work out the value of $\left(\frac{16}{81}\right)^{\frac{3}{4}}$

Fourth roots: $2^4 = 16$ and $3^4 = 81$.

1 mark
Take the fourth root, then cube

$\frac{2^3}{3^3} = \frac{8}{27}$

1 mark
Exact fraction

$\frac{8}{27}$

(2)

$3^a = \frac{1}{9}$ $\quad 3^b = 9\sqrt{3}$ $\quad 3^c = \frac{1}{\sqrt{3}}$

(b) Work out the value of $a + b + c$

$3^2 = 9$, so $1 \div 9 = 3^{-2}$ and $a = -2$.

$9\sqrt{3} = 3^2 \times 3^{0.5}$ so $b = 2 + 0.5 = 2.5$

$1 \div \sqrt{3} = 3^{-0.5}$ so $c = -0.5$

1 mark
Write each number as a power of 3

$a + b + c = -2 + 2.5 - 0.5 = 0$

1 mark
Sum the three exponents

0

(2)

(Total for Question 14 is 4 marks)